BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Tracing Programming Constructs

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. When is a condition-controlled loop suitable?

2. Linear-search [7, 2, 9, 4] for 4. How many comparisons are needed?

A) 1 B) 2
C) 4 D) 3

3. "An IF statement must execute every possible branch regardless of its condition." Correct this. Explain why.

4. Do one bubble-sort pass from left to right, smallest to largest, on [6, 1, 5, 3]. Show the array after each comparison. Which value ends up on the right?

Today's objectives

Trace programs using sequence, selection and iteration. Apply the programming techniques to a new problem.

Key vocabulary - Read, cover, recall

Trace table
Records the current variable values after relevant instructions.

Selection tracing
Only the true branch of an IF/ELSE is followed.

Iteration tracing
Record the changes for every repeated execution.

Debugging
Compare expected and traced results to find the first incorrect step.

Write one key term and its meaning in your own words:

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Learn and practise

New learning - Key ideas

Trace table	Records the current variable values after relevant instructions. For example, consider a score calculator.
Selection tracing	Only the true branch of an IF/ELSE is followed. For example, consider a discount total.
Iteration tracing	Record the changes for every repeated execution. For example, consider a lesson attendance.
Debugging	Compare expected and traced results to find the first incorrect step. For example, consider a countdown.

I Do - One worked example

A score calculator is being developed. Explain how trace table and selection tracing are used. [4 marks]

1. Identify where this particular program needs control or values.

2. Trace table [1]; Records the current variable values after relevant instructions. [1].

3. Selection tracing [1]; Only the true branch of an IF/ELSE is followed. [1].

4. Give one accurate point and a linked reason for each technique.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. A discount total is being programmed. Explain why trace table matters. [2 marks]

2. For a lesson attendance, explain how selection tracing is used. [2 marks]

3. A countdown needs iteration tracing and debugging. Explain the roles of both. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Predict the value printed after the loop.

```
totalGB = 0
for i = 1 to 3
totalGB = totalGB + i
next i
print(totalGB)
```

BLACK PEN: predict or complete the code; test, then improve.

You Do - Independent exam questions

Q2. State TWO concepts needed for this programming technique. [2 marks]

Q3. Explain how selection tracing and iteration tracing can help a points accumulator. [4 marks]

Q4. Explain the roles of trace table, iteration tracing and debugging when creating a fitness tracker. [6 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. A trace table records the values of variables after relevant instructions execute. ☐ T ☐ F
2. Records the current variable values after relevant instructions ☐ T ☐ F
3. Only the true branch of an IF/ELSE is followed ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 When repetition continues or ends according to a condition, such as valid input. ☐

Q2 C - 4 ☐

Q3 It chooses a branch using a condition. Selection can be nested. ☐

Q4 (a) [[1, 6, 5, 3], [1, 5, 6, 3], [1, 5, 3, 6]]; largest value 6 is at the right end. ☐

Code First - Model

```
totalGB = 0
for i = 1 to 3
totalGB = totalGB + i
next i
print(totalGB)
```

Output: 6

We Do - Check the model

1. Trace table [1]; Records the current variable values after relevant instructions. [1]. ☐

2. Selection tracing [1]; Only the true branch of an IF/ELSE is followed. [1]. ☐

3. Iteration tracing [1]; Record the changes for every repeated execution. [1]. Debugging [1]; Compare expected and traced results to find the first incorrect step. [1]. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Trace table [1]; Iteration tracing [1]. ☐

Q3 Selection tracing [1]; Only the true branch of an IF/ELSE is followed. [1]. Iteration tracing [1]; Record the changes for every repeated execution. [1]. ☐

Q4 Trace table [1]; Records the current variable values after relevant instructions. [1]. Iteration tracing [1]; Record the changes for every repeated execution. [1]. Debugging [1]; Compare expected and traced results to find the first incorrect step. [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: